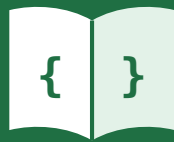

Spring Boot 4 — From the Basics to Production

Course Syllabus

Spring Boot Edu



2026-09-25

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1. About the Course

A hands-on course on Spring Boot 4.1 and Java 27 in 25 modules and a capstone project. Every module is a small, runnable Maven project around the same domain — a bookstore (Book, Author, Customer, Order, Review) — so that you learn Spring, not a new domain every week.

Every module contains:

Part	What it is
lesson/	runnable examples with tests — always green
docs/	lesson notes and exercises in Turkish and English (Markdown and PDF)
exercise/	starter code with TODOs — its tests are red until you solve them
solution/	the reference solution with the same tests — green
README.md, requests.http	how to run the module, HTTP examples for the IDE

Audience: developers who know Java and are new to Spring, or who used Spring before and want the state of 2026 (Spring Boot 4, Spring Framework 7, Java 27).

Format: self-study or classroom. A week of the plan below is about 9 hours: lesson notes and examples ($\approx 60\%$), exercises ($\approx 40\%$).

2. Prerequisites and Setup

Knowledge:

- Java: classes, interfaces, generics, collections, lambdas and streams
- basic SQL (SELECT, JOIN, INSERT) and HTTP (methods, status codes, JSON)
- Git and a terminal

Machine:

Tool	Needed for
JDK 27	everything (the Maven wrapper brings Maven 3.9)

Tool	Needed for
Docker with ≥ 8 GB memory	databases and brokers from module 05 on (Testcontainers, Docker Compose)
an IDE (IntelliJ IDEA, VS Code)	requests.http files, debugging
kind, kubectl, Helm	modules 22 and 23, the capstone on Kubernetes
≈ 2 GB free disk for models	module 24 (Ollama)

Module 00 walks through the setup; `./mvnw verify` in the repository root proves that everything works.

3. Learning Outcomes

After the course you will be able to:

1. build REST, GraphQL, WebSocket and gRPC APIs with Spring Boot 4, validated and with RFC 9457 error answers
2. store data in PostgreSQL (JDBC, JPA), MongoDB, Redis, Hazelcast and Elasticsearch — and choose between them
3. connect services reliably: HTTP clients with retries, Kafka with a transactional outbox and idempotent consumers
4. secure applications with Spring Security: form login, JWT resource servers, OAuth2 and method security
5. test on every level: unit tests, slice tests, Testcontainers, contract fakes, end-to-end tests
6. make applications observable (metrics, traces, logs with OpenTelemetry and Grafana) and production-ready (Docker images, native images, Kubernetes, Helm)
7. structure larger applications with Spring Modulith and Spring Cloud, and add AI features with Spring AI
8. put all of it together in a system of several services (capstone)

4. Weekly Plan

The order follows the module numbers; each module builds on the ones before it (see section 5).

Week	Modules	Main topics	Hours
1	00, 01	setup, modern Java 21 → 27; IoC and dependency injection	7
2	02, 03	configuration, auto-configuration, a starter; REST with Web MVC	9
3	04, 05	RestClient, HTTP interfaces, @Retryable; JDBC, Flyway, Testcontainers	9
4	06, 07	JPA and Hibernate; MongoDB	11
5	08, 09	Redis and caching; Hazelcast (maps, locks, entry processors)	8
6	10, 11	Elasticsearch; Kafka, outbox, Kafka Streams	11
7	12, 13	Spring Security, JWT, OAuth2; reactive programming, WebFlux, R2DBC	12
8	14, 15	testing strategies; observability with OpenTelemetry and Grafana	10
9	16, 17	async, scheduling, Spring Batch; GraphQL and WebSocket	10

Week	Modules	Main topics	Hours
10	18, 19	Spring Modulith; AOT processing, the JVM AOT cache, GraalVM native image	8
11	20, 21	Docker images, jlink, Buildpacks; gRPC	8
12	22, 23	Kubernetes with kind; Spring Cloud (Gateway, Config, LoadBalancer, Circuit Breaker)	11
13	24, capstone	Spring AI (chat, structured output, RAG, tools, MCP); start the capstone	5 + 4
14	capstone	the bookstore platform: guide, exercises, Kubernetes	6–8

★ Tip

Short on time? Weeks 1–8 are the core of everyday Spring work. Modules 16–24 can be taken in any order once their prerequisites are done.

5. Module Overview

5.1 Foundations

Module	Topic	Prerequisites	Hours
00	Setup and modern Java (21 → 27)	basic Java	3
01	Spring core container: IoC and dependency injection	00	4
02	Configuration and auto-configuration	01	4
03	REST APIs with Web MVC	01, 02	5

Module	Topic	Prerequisites	Hours
04	HTTP clients and resilience	01-03	4

5.2 Data

Module	Topic	Prerequisites	Hours
05	Spring JDBC and PostgreSQL	01-03, basic SQL	5
06	Spring Data JPA and Hibernate	05	6
07	Spring Data MongoDB	05, 06	5
08	Redis and caching	01-03	4
09	Distributed data with Hazelcast	08	4
10	Search with Elasticsearch	06	5

5.3 Integration and Security

Module	Topic	Prerequisites	Hours
11	Messaging with Kafka	06	6
12	Security with Spring Security	03, 06	6
13	Reactive programming and WebFlux	03, 07	6
14	Testing Spring Boot applications	06	5
15	Observability: metrics, traces, logs	03	5
16	Async, scheduling and Spring Batch	06	5
17	GraphQL and WebSocket	06, 13	5
18	Spring Modulith	06, 11	4

5.4 Production and Beyond

Module	Topic	Prerequisites	Hours
19	Native image and performance	03, 05	4

Module	Topic	Prerequisites	Hours
20	Docker and deployment	15	4
21	gRPC	03	4
22	Kubernetes	20	5
23	Spring Cloud	04, 15, 22	6
24	Spring AI	06	5
capstone	Bookstore platform: 4 services + gateway, all technologies together	all	8-12

6. Assessment

- **Exercises:** at least three per module. An exercise is done when its tests are green — the tests are the acceptance criteria, and they are the same tests as in the solution. Each exercise names its estimated time and difficulty.
- **Capstone:** three exercises on the running platform (idempotency, cancellation with compensation, retries) plus optional challenges. The end-to-end test must stay green.
- **Suggested grading for a classroom course:** exercises 50 %, capstone exercises 30 %, a short presentation of one extra challenge 20 %.

7. Technology Stack

Area	Version
Java	27
Spring Boot	4.1.1 (Spring Framework 7.0, Spring Security 7.1, Spring Data 2026.0)
Spring Cloud / Spring AI / Spring Modulith	2025.1.3 / 2.0.1 / 2.1.1
Build and test	Maven 3.9 (wrapper), JUnit 6, AssertJ, Mockito, Testcontainers 2

Area	Version
Infrastructure (Docker)	PostgreSQL 18, MongoDB 8, Elasticsearch 9, Redis 8, Kafka 4 (KRaft), Hazelcast 5.5, Ollama, Grafana LGTM
Deployment	Docker (jlink images), GraalVM native image, kind, Kustomize, Helm

8. Tips for Learners and Teachers

- Run the lesson first (`./mvnw -pl modules/NN-.../lesson spring-boot:run`), then read the notes next to the code: every example names its lesson section.
- Do the exercises before reading the solution. The hints in each exercise point to the lesson section that helps.
- The “Common Mistakes” section of every lesson is a good starting point for classroom discussions.
- For a classroom course, start each week with the architecture of the module (section 2 of the lesson) and use the `requests.http` file for live demos.